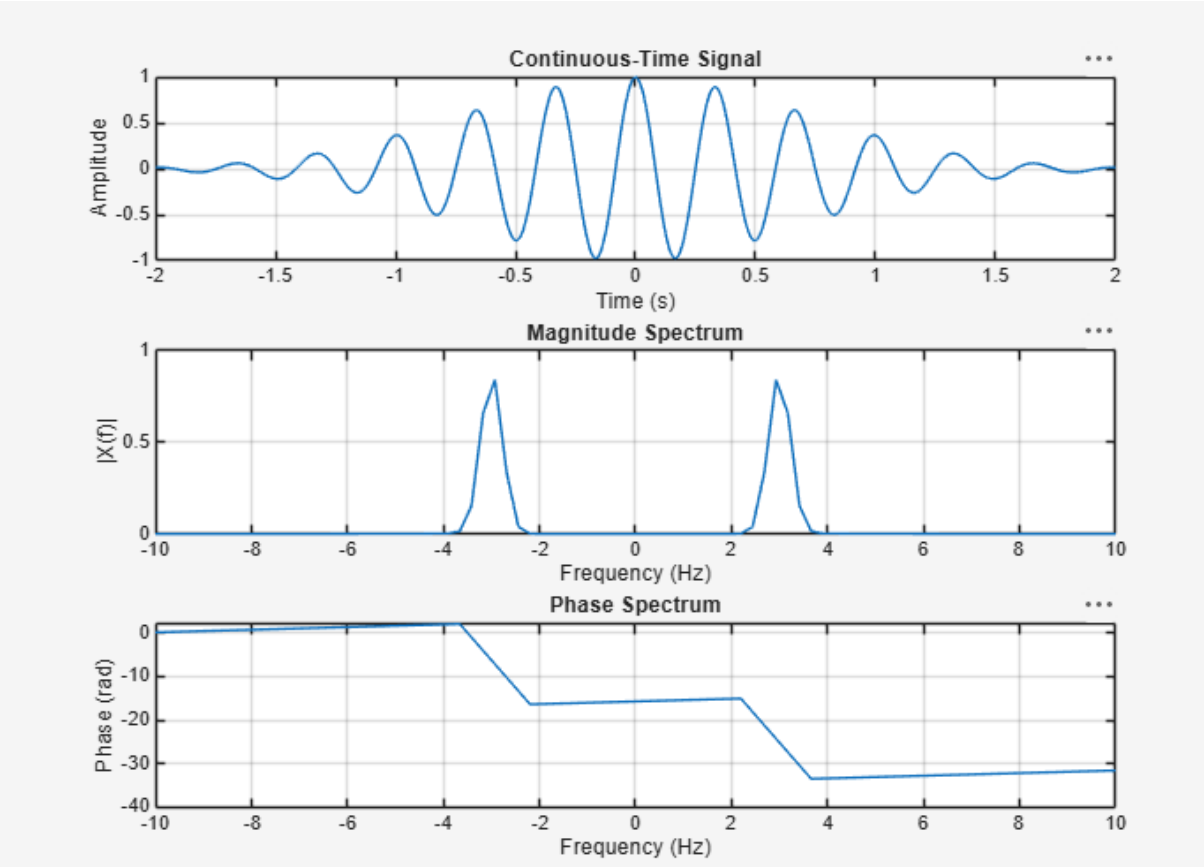
Experiment – 7:
Continuous-Time Fourier Transform and Spectrum

(ii)

/MATLAB Drive/Experiment_7_2.m

```
clc; clear; close all;
%% 7(ii) Determination of Magnitude and Phase Spectra of Continuous-Time Signals
figure;
% Magnitude and Phase Spectrum of a CT Signal
t = -2:0.001:2;
dt = t(2)-t(1);
x = exp(-t.^2).*cos(2*pi*3*t);
N = 4096;
X = fftshift(fft(x,N))*dt;
f = (-N/2:N/2-1)/(N*dt);
magX = abs(X);
phaseX = unwrap(angle(X));
subplot(3,1,1);
plot(t,x); grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Continuous-Time Signal');
subplot(3,1,2);
plot(f,magX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('|X(f)|');
title('Magnitude Spectrum');
subplot(3,1,3);
plot(f,phaseX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
title('Phase Spectrum');
```

Code



Output